

Measuring Correlation Risk for Energy Derivatives

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6.1 INTRODUCTION

The importance of correlation as a measure of dependence has been often emphasized in the context of pricing derivatives whose payoffs depend on the joint distribution of underlying prices, indices, or rates. There is no doubt that at present these derivatives are getting noticeably more popular and numerous. For example, they include a vast class of basket options, i.e. options on linear combinations of various price indices from different markets. Another example can be found in commodity markets, where spread options, both standard and, increasingly, “multi-legged”, are omnipresent.

Obviously, there are limitations to the use of the correlation, especially in the case of complex joint distributions. However, even under these circumstances practical considerations often force one to use correlations as measure of dependence.

Once the correlation is used for pricing, an immediate question arises: namely, the question of estimating the sensitivity of derivative prices to the correlation parameters, and with it the question of corresponding correlation risk measure. In this chapter we discuss various ways to define and compute one such measure, the correlation VaR.

We start with a brief recap of the correlation and its properties. We discuss various approaches to parametrization of the correlation matrix. We then introduce different methods to generate distributions of correlations matrices. Finally, we apply these methods to determine the correlation VaR of a typical energy derivative. We conclude with a discussion.

6.2 CORRELATION

Linear correlation is the most widely used measure of dependence between random variable X and Y with finite variances. It is defined as:

$$\rho(X, Y) = \frac{\text{Cov}[X, Y]}{\sqrt{\sigma^2[X]\sigma^2[Y]}}$$

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Risk Management in Commodity Markets: From Shipping to Agriculturals and Energy Edited by Hélyette Geman

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where $\text{Cov}[X, Y]$ is the covariance between X and Y , $\text{Cov}[X, Y] \equiv E[XY] - E[X]E[Y]$ and $\sigma^2[X] \equiv \text{Cov}[X, X]$, $\sigma^2[Y]$ denote the variances of X and Y . The linear correlation is a measure of linear dependence. For the *elliptical* family of joint distributions, which include joint normal, lognormal and others, the linear correlation together with the variances is sufficient to describe the dependency structure. The pitfalls and limitations of the concept of a linear correlation are investigated in detail in the excellent paper by Embrechts, McNeil and Straumann (1999).

If we are dealing with n random variables $X_1, X_2, \dots, X_n$ (for example returns of financial variables), then correlations ρ_{ij} between different pairs i, j of returns are expressed in terms of matrices. Correlation matrices must satisfy the following properties ($i, j = 1, \dots, n$):

- All entries have to be in the interval $[-1, 1]$: $-1 \leq \rho_{ij} \leq 1$
- The diagonal terms of a correlation matrix are equal to one: $\rho_{ii} = 1$
- The matrix has to be symmetric: $\rho_{ij} = \rho_{ji}$
- The matrix has to be positive *semidefinite*, i.e., the variance of any portfolio with correlation matrix ρ , is non-negative: $\forall W, V \sigma_p^2 = (Z)^T \rho(Z) Z \geq 0$, where $W = w_1, \dots, w_n$ is the array of weights of the portfolio, $V = v_1, \dots, v_n$ is the array of standard deviations of the returns, $Z = w_1 v_1, \dots, w_n v_n$ and T denotes transpose of a matrix.

In financial applications, correlation matrices are most commonly estimated from the historical data. Another way is to get *implied* correlations, i.e. to calibrate a model using market prices of correlation-dependent derivatives in the same way implied volatilities are obtained from options quotes. The advantage of the first approach is that typically more data is available for its implementation. The drawback—it is backward looking. The second approach has less data, but it is forward looking.

6.3 PERTURBING THE CORRELATION MATRIX

In order to estimate correlation risk we need an ability to generate a random sample of correlation matrices, so that we can analyze the corresponding distribution of portfolio values, i.e. calculate the *correlation VaR*. This task is not easy, since a correlation matrix is a fairly rigid object, especially with respect to the requirement of positive definiteness.

In this section we will discuss four different approaches to perturbing the correlation matrix: bootstrapping, element-wise perturbation, perturbation with the help of angle parametrization of the Cholesky matrix, and perturbation of the eigen-values.

6.3.1 Bootstrap method

The bootstrap methodology is a Monte Carlo *resampling* method. It was introduced by Efron (1979). Suppose that we want to estimate the density of the correlation ρ between two random variables X and Y based on historical observations $\{X_i, Y_i\}_{i=1}^N$. The basic bootstrapping algorithm to create a distribution for ρ is as follows.

1. Generate N uniform i.i.d. random integers $n_1, n_2, \dots, n_N$ in the interval $[1, N]$.
2. Create a sample $\hat{X} \equiv \{X_{n_i}, Y_{n_i}\}_{i=1}^N$ using the $n_1, \dots, n_N$.
3. Calculate $\hat{\rho}$ using the sample $\hat{X}$.
4. Repeat the previous steps M times with M being a large number.

Instead of drawing single values from observations, one can use *block* bootstrap. This means that the data is split into blocks of length m and the above procedure is applied. Fengler and Schwendner (2003) used blocks of length $m = 3$. They quantify the error in the correlation estimates from historical data by approximating the asymptotic distribution of the correlation via block bootstrapping. The bootstrapped correlation distributions are then mapped on prices of three standard types of multi-asset options: a basket option, minimum option and maximum option. Bid-ask spreads of the prices are computed at statistical quantiles of the resulting price distributions.

6.3.2 Perturbing individual correlations

Turkay *et al.* (2003) propose perturbing the correlation matrix locally to a desired target matrix while ensuring that the irrelevant correlations remain the same, and the new correlation matrix remains positive semi-definite. They obtain an analytical solution for the bounds of a single correlation term of a positive semi-definite correlation matrix. The methodology is based on the elegant idea of re-ordering of the assets that define the matrix and applying the Cholesky decomposition to localize the perturbation to the last entries in the Cholesky matrix. They also present an iterative application of the single correlation stress test methodology in order to stress a number of correlations.

6.3.3 Perturbing angles in the angle representation of the correlation matrix

The *Triangular Angles Parametrization* (TAP) parametrization of the correlation matrix through a *unique* set of angles was put forward in Brigo *et al.* (2002). This is an extension of results proposed in Rebonato and Jäckel (2000), which were the first to apply the results of Pinheiro and Bates (1996) in a financial context. See also Geman and Souveton (1997).

The essential idea is that the correlation matrix can be parameterized through a unique lower-triangular matrix, where the entries are angles taking values in $[0, \pi]$. This angle-representation maps to a Cholesky matrix from which we can compute the correlation matrix. The good thing about this approach is that we automatically satisfy the correlation constraints specified in Section 6.2.

We start with the Cholesky decomposition of the correlation matrix (see Section 6.2). In this decomposition the correlation matrix ρ is represented as a product of a lower-triangular matrix L and its transpose:

$$\rho_{ij} = \sum_{k=1}^N l_{ik} l_{jk}$$

The elements of the lower-triangular matrix L are then parameterized in terms of cosines and sines of $N(N - 1)/2$ angles θ_{ij} ($j < i$) between 0 and π .

$$l_{ij} = \begin{cases} \cos \theta_{ij} \prod_{k=1}^{j-1} \sin \theta_{ik} & j < i \\ \prod_{k=1}^{i-1} \sin \theta_{ik} & j = i \end{cases} \quad (6.1)$$

The angles are found via a robust and efficient procedure which makes the whole approach very attractive. Moreover, through these angles the space of *all* correlation matrices is covered.

One implication of having the angle representation of correlation matrices is that if we have an algorithm to sample the angles, then we also have an algorithm to sample the correlations with some width around the base correlations, thus providing an alternative way to generate the correlation VaR.

The simplest algorithm for generating distribution of correlation matrices is to generate random angles θ_{ij} around the base angles θ_{ij}^0 with some distribution $\pi(\theta_{ij}|\theta_{ij}^0)$, which is symmetric and centered around the base-correlation θ_{ij}^0 for every i, j . Random angles are simulated using the historical distribution. From our analysis we find that the angles tend to be distributed around the mean, with a standard deviation in the order of $\sigma = 5\%$. Based on this analysis we propose two perturbation methods. In the first approach we perturb all angles using one standard normal variate $z \sim N(0, 1)$,

$$\hat{\theta}_{ij} = \arctan(\tan(\theta_{ij} + \frac{\pi}{2})(1 + \sigma z)) + \frac{\pi}{2} \quad (6.2)$$

In the second approach all angles θ_{ij} are perturbed by i.i.d. standard normal variates $z_{ij} \sim N(0, 1)$,

$$\hat{\theta}_{ij} = \arctan(\tan(\theta_{ij} + \frac{\pi}{2})(1 + \sigma z_{ij})) + \frac{\pi}{2} \quad (6.3)$$

6.3.4 Perturbing eigenvalues

In this section we consider the generation of random correlation matrices around the base correlation matrix through the perturbation of eigenvalues. We can express the correlation matrix in terms of its eigen-system (Λ, V) via

$$\rho_{ij} = \sum_{k,l=1}^n V_{ik} \Lambda_{kl} V_{lj}$$

where $\Lambda_{kl} \equiv \lambda_k \delta_{kl}$ is the diagonal matrix with the eigenvalues λ_i on the diagonal. We assume that the eigenvalues by $\lambda_1 \geq \lambda_2 \geq \dots \lambda_n \geq 0$. Furthermore the eigenvalues satisfy the constraint $\sum_{k=1}^n \lambda_k = n$.

We consider the following four algorithms to perturb the eigenvalues. The first algorithm is as follows.

1. Generate n i.i.d random standard normal variates $z_i \sim N(0, \sigma_i)$, ($i = 1, \dots, n$)
2. Compute the perturbed eigenvalues

$$\hat{\lambda}_i = \lambda_i e^{\sigma_i z_i}$$

The other three algorithms are variations of the following algorithm.

1. Generate a random index $K \in [1, \dots, n]$ according to some distribution $p_i \geq 0$ with $\sum_{i=1}^n p_i = 1$.
2. Generate an i.i.d. standard normal variable $z \sim N(0, 1)$.

3. Compute the perturbed eigenvalue λ_K

$$\hat{\lambda}_K = \lambda_K e^{\sigma_K z}$$

The three cases we will consider are

1. Perturb the largest eigenvalues more: $p_i = \frac{\lambda_i}{n}$
2. Perturb the eigenvalues uniformly: $p_i = \frac{1}{n}$
3. Pick one specific eigenvalue: $p_i = \delta_{iK}$

Having perturbed eigenvalues, we define the perturbed correlation matrix via

$$\hat{\rho}_{ij} = \sum_{k,l=1}^n V_{ik} \hat{\Lambda}_{kl} V_{lj}$$

where $\hat{\Lambda}_{kl} \equiv \hat{\lambda}_k \delta_{kl}$ is the diagonal matrix with the perturbed eigenvalues $\hat{\lambda}_i$. To ensure that $\sum_{i=1}^n \rho_{ii} = n$ we need to renormalize the random correlation matrix $\hat{\rho}$:

$$\tilde{\rho}_{ij} = \frac{\hat{\rho}_{ij}}{\sqrt{\hat{\rho}_{ii} \hat{\rho}_{jj}}}$$

6.4 CORRELATION VAR

The question we want to ask is the sensitivity of the value of a trade or portfolio to the correlation matrix. To this end we need an efficient and practical method to perturb the correlation matrix. We will use the methods discussed in the previous section to compute the sensitivity of a portfolio to correlation.

As we mentioned above, how we perturb the correlations, i.e. what probabilities we attach to a particular perturbation of the correlation matrix, is an important and difficult choice. We will take a practical point of view here and leave the more satisfying theoretical results for a future work.

In calculating the correlation VaR, we will perturb the correlation matrix according to some perturbation scheme, which effectively means that we have some distribution $\pi(\rho|\rho^0)$ and compute the density for the portfolio value as a function of the correlation as follows.

$$\pi(v) = \int \delta(v - V(\rho)) \pi(\rho|\rho^0) d\rho \quad (6.4)$$

From this we can then determine the correlation VaR.

6.5 SOME EXAMPLES

In this section we apply the above methodology to the valuation of a typical energy derivative subject to correlation inputs. We value a hypothetical power plant with a switching capability

between the gas and oil. The value of the power plant is modeled by the following strip of spread options:

$$V = Q \left\{ \sum_{m=1}^{m=N} H_m^{\text{on}} S_m^{\text{on}} + H_m^{\text{off}} S_m^{\text{off}} \right\} \quad (6.5)$$

where H_m^{on} and H_m^{off} are the numbers of on-peak and off-peak hours in month m and S_m^{on} and S_m^{off} are the spread options for month m , whose payoff for month m is given by

$$S_m^{\text{on}} = \max \{ P^{\text{on}} - \text{HR}_G G, P^{\text{on}} - \text{HR}_F F, 0 \} \quad (6.6)$$

$$S_m^{\text{off}} = \max \{ P^{\text{off}} - \text{HR}_G G, P^{\text{off}} - \text{HR}_F F, 0 \} \quad (6.7)$$

Here P^{on} and P^{off} are the on-peak and off-peak power prices, G is the gas price, F is the oil price and HR_G , HR_F are the heat rates for gas and oil respectively. For simplicity we ignore variable costs and assume zero interest rates.

We value the power plant on 1 January 2006, over the period of calendar year 2007. The inputs listed in Table 6.2 include forward on-peak and off-peak prices, gas and oil forward prices, volatilities for on-peak, off-peak power, gas and oil. For the current example we have $N = 12$ and the amount Q of MW is set to $Q = 100$.

For a given correlation matrix ρ we compute the value of the power plant using a Monte Carlo simulation. We repeat this for a sample of random correlation matrices using one of the methods described in the previous sections. Using Eq. (6.4), we then compute the density of the value V in Eq. (6.5). In order to get a distribution of the eigenvalues or the angles, we perform a bootstrap on historical prices. Based on this analysis, we make the assumption that the eigenvalues have a lognormal distribution and generate the random angles using Eq. (6.3).

The base correlation matrix ρ_{ij}^0 is calculated from historical returns of forward prices. We use three years of historical data. Given the base correlation matrix we can compute the corresponding Cholesky decomposition and the lower-triangular angle matrix θ_{ij} . In order to estimate the variations of the angles, we compute the angles using a sliding window. We calculated the correlation matrix and the corresponding angle matrix for every year using that year's data. In general we find that the standard deviation of the angles is in the order of 5 %. We will use this number to perturb the angles.

We apply the methods we discussed earlier to simulate correlation matrices and compute the density of the power plant values as a function of the correlation perturbations. In Table 6.1 we give the mean, standard deviation, and 95 % left VaR for the power plant value as a function of the different methods that we discussed in this chapter. We use 1000 simulations of correlation matrices, and 10 000 simulations in the underlying Monte Carlo to evaluate the V . The corresponding histograms are in Fig. 6.1. From these histograms it is clear that the distribution of the basket values depends on the method of perturbing the correlation matrix. If we choose a uniform density over the eigenvalues the smallest eigenvalues gets too much weight and the density is very narrow. As we put more weight on the larger eigenvalues the density will widen. The case where we only perturb the largest eigenvalue can be considered as the worst-case scenario. As for the perturbation through the angles we see that, as expected, the case with a *correlated* perturbation of the angles leads to a wider distribution of the value.

Table 6.1 Power plant with value 15.8MM. Sample size is 1000

Method	Largest eigenvalue	Weighted eigenvalues	Uniform eigenvalues
Mean (MM)	15.81	15.80	15.81
Standard deviation (MM)	0.23	0.17	0.09
95 % CorVaR (MM)	15.41	15.53	15.66

Method	Parallel shift angles	Uncorrelated angles	Bootstrap
Mean (MM)	15.79	15.82	16.04
Standard deviation (MM)	0.67	0.22	0.65
95 % CorVaR (MM)	14.69	15.47	15.01

Table 6.2 Forward prices and volatilities inputs for the valuation of a power plant

Contracts	Power on-peak	Power off-peak	Gas	Oil	On peak vol	Off peak vol	Gas vol	Oil vol
1/1/2007	100	70	10.2	50.5	0.83	0.71	0.64	0.27
2/1/2007	90	65	9.4	48	0.69	0.59	0.67	0.27
3/1/2007	80	60	8.4	50	0.65	0.55	0.69	0.26
4/1/2007	80	56	8.2	50	0.50	0.43	0.42	0.26
5/1/2007	75	50	7.5	51	0.52	0.44	0.40	0.26
6/1/2007	85	54	6.8	50	0.54	0.45	0.40	0.26
7/1/2007	110	70	6.5	49	0.58	0.49	0.40	0.25
8/1/2007	105	70	6.4	50	0.58	0.49	0.40	0.25
9/1/2007	90	60	6.6	49	0.52	0.44	0.42	0.25
10/1/2007	85	61	7	49.2	0.51	0.43	0.44	0.24
11/1/2007	93	65	9	49	0.51	0.43	0.48	0.24
12/1/2007	102	78	10.8	49.5	0.51	0.43	0.47	0.24

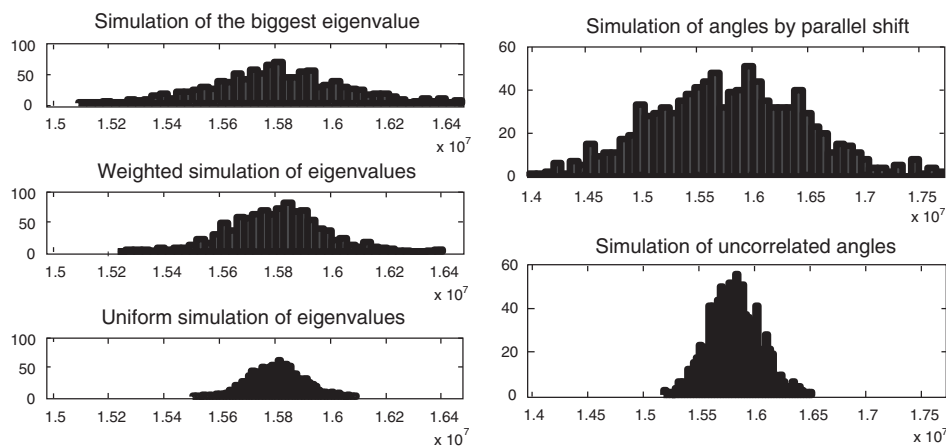


Figure 6.1 We plot the histograms for the value of the power plant with all different methods discussed in the article. The sample size is 1000

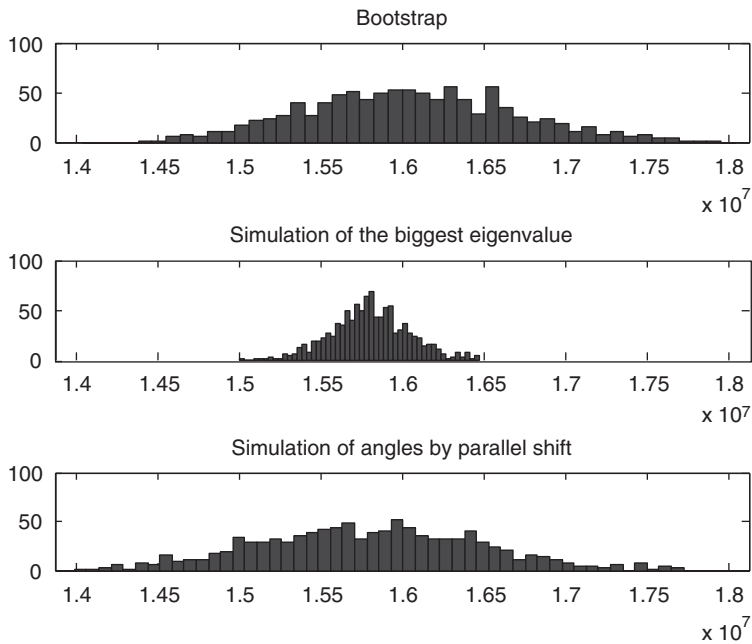


Figure 6.2 Comparison of the three different cases: bootstrap, largest eigenvalue, parallel shift angles. The graph shows the results for the power plant value

In Fig. 6.2 we compare the bootstrap method with the largest eigenvalue perturbation and the parallel shift of angles for the two examples we consider. It shows that the distribution of the bootstrap case is wider than the case of the other two methods.

6.6 DISCUSSION AND CONCLUSIONS

We have described various methods to perturb the correlation matrix and compute the correlation VaR. The simplest method is the bootstrap method, which is essentially a resampling of the historical timeseries. The second method involves the local perturbation of the correlation matrix elements. The third method uses a perturbation of the correlation matrix through perturbation of the angles in the TAP parametrization. Finally we propose a method to perturb the correlation matrix via the perturbation of the eigenvalues.

We provide some numerical results for these methods applied to the valuation of a power plant with two fuels. Different methods lead to different results for the correlation VaR. This is not surprising since there is clearly a lot of freedom in the choice of perturbations and underlying densities. Given this freedom, it would be natural to argue in favor of using the simplest method, the bootstrapping of the correlation matrix. However, there is a clear advantage in having a perturbation method with a readily available control parameter. Among other benefits, such a method is very useful for devising and carrying out stress testing. Needless to say, both the angle and eigenvalue perturbation methods do provide such control parameters.

To summarize, the present work shows that the use of some parameterized method, be it angles or eigenvalues, gives us a convenient tool to stress-test complicated portfolios of

instruments and to do it in a well-defined manner. A consistent application of the correlation VaR measures described in this chapter improves our understanding of product sensitivity to correlation and provides us with a useful approach to product comparison.¹

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